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Inventor(s)

IIJIMA; ICHIRO et al.

AUTHENTICATION APPARATUS, AUTHENTICATION TARGET APPARATUS, IMAGE FORMING APPARATUS, AND REPLACEMENT UNIT FOR THE IMAGE FORMING APPARATUS

Abstract

An authentication apparatus that authenticates an authentication target apparatus, includes: an acquisition unit configured to acquire, from the authentication target apparatus, information indicating secret data stored in the authentication target apparatus, a first hash value that corresponds to the secret data, and a repeat count N, wherein N is an integer of 2 or more; an operation unit configured to obtain a second hash value from input data that is based on the secret data, by performing N repeated operations using a one-way function; and an authentication unit configured to authenticate the authentication target apparatus by comparing the first hash value acquired by the acquisition unit from the authentication target apparatus with the second hash value obtained by the operation unit.

Inventors: IIJIMA; ICHIRO (Kanagawa, JP), ENOMOTO; HISASHI (Shizuoka, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

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Background/Summary

BACKGROUND

Field of the Disclosure

[0001] The present disclosure relates to a technology for authentication of an authentication target apparatus by an authentication apparatus.

Description of the Related Art

[0002] If a component that is not expected in the design is connected to an apparatus, this may cause a failure of the apparatus or an unexpected event. Therefore, it is necessary to determine whether or not the component attached to the apparatus is a genuine product, and to alert the user if the component is determined to be not a genuine product. International Publication No.

2018/080497 discloses a configuration that uses a password, a hash value, and an encryption key to determine whether or not a replacement unit attached to an apparatus is a genuine product.

Typically, the larger the data lengths (bit lengths) of the password, hash value, and encryption key are, the higher the resistance to cryptanalysis attacks such as brute-force attacks can be.

[0003] However, due to an improvement in processing capacity of computers, the data length required to maintain sufficient resistance to cryptanalysis attacks has increased, and is expected to continue to increase in the future. On the other hand, memory devices that store information for use in authentication may have limited capacity due to the cost and the like. Therefore, the data length of information for use in authentication may not be sufficient to resist cryptanalysis attacks.

SUMMARY

[0004] According to a present disclosure, an authentication apparatus that authenticates an authentication target apparatus, includes: an acquisition unit configured to acquire, from the authentication target apparatus, information indicating secret data stored in the authentication target apparatus, a first hash value that corresponds to the secret data, and a repeat count N, wherein N is an integer of 2 or more; an operation unit configured to obtain a second hash value from input data that is based on the secret data, by performing N repeated operations using a one-way function; and an authentication unit configured to authenticate the authentication target apparatus by comparing the first hash value acquired by the acquisition unit from the authentication target apparatus with the second hash value obtained by the operation unit.

[0005] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a diagram showing a configuration of an image forming apparatus according to one or more aspects of the present disclosure.

[0007] FIG. 2 is a diagram showing a configuration of an authentication system according to one or more aspects of the present disclosure.

[0008] FIG. 3A is a diagram illustrating a relationship between information of a secret data table and information of a hash value table.

[0009] FIG. 3B is a diagram illustrating a relationship between information of the hash value table

and information of a combined hash value.

[0010] FIG. 4 is a flowchart of authentication processing according to one or more aspects of the present disclosure.

[0011] FIG. 5 is a diagram showing a configuration of an authentication system according to one or more aspects of the present disclosure.

[0012] FIG. 6 is a flowchart of processing that is executed by an authentication target chip in the authentication processing.

DESCRIPTION OF THE EMBODIMENTS

[0013] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed disclosure. Multiple features are described in the embodiments, but limitation is not made to an disclosure that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

[0014] FIG. 1 is a diagram showing an image forming apparatus **100** that constitutes an authentication system according to the present embodiment. The image forming apparatus **100** is, for example, a printer, copier, or multifunctional machine, and forms images on sheets according to an electrophotographic process. A process cartridge **102** is a unit (replacement unit) that can be detached from the body of the image forming apparatus **100**. The process cartridge **102** includes, for example, a photoreceptor and members required to form a toner image on the photoreceptor. The process cartridge **102** performs part of processing of forming an image on a sheet under the control of an engine controller **101**. The process cartridge **102** includes an authentication target chip **103**. The authentication target chip **103** is a tamper-resistant device, and functions as an authentication target apparatus.

[0015] The engine controller **101** provided in the main body of the image forming apparatus **100** includes a control unit **105**, an authentication chip **106**, a communication circuit **104**, and an attachment sensor **107**, and functions as an authentication apparatus that authenticates the authentication target apparatus. The control unit **105** serves also as a controller that comprehensively controls the entire image forming apparatus **100**. The authentication chip **106** performs processing of authenticating the authentication target chip **103**. The authentication chip **106** may be a tamper-resistant device. The communication circuit **104** performs communication processing with the process cartridge **102**. The attachment sensor **107** detects the attachment of the process cartridge **102** to the main body of the image forming apparatus **100**. When the attachment sensor **107** detects the attachment of the process cartridge **102**, the control unit **105** starts authentication processing, which will be described later.

[0016] FIG. 2 is a diagram showing a configuration of the authentication system including the authentication chip **106** and the authentication target chip **103**. An input/output circuit **201a** of the authentication chip **106** relays data that is transmitted and received between the control unit **105** and a microcomputer **202a**. The microcomputer **202a** executes necessary processing in accordance with commands from the control unit **105**. At this time, the microcomputer **202a** stores data necessary for the processing in a volatile memory **203a**. A non-volatile memory **204a** stores a control program **205a** for the authentication chip **106**, a signature verification key **215**, and a chip identifier table **216**. The chip identifier table **216** stores identifiers (chip identifiers) of authentication target chips **103** that were successfully authenticated in the past. A verifier **206** verifies a digital signature based on the signature verification key **215**. An operation unit **207a** uses a one-way function F to obtain hash values of input data. A comparator **208** determines whether or not two pieces of input data match each other. Note that a configuration is also possible in which the microcomputer **202a** performs the processing performed by one or more of the verifier **206**, the operation unit **207a**, and the comparator **208**.

[0017] The image forming apparatus **100** is configured so that, when the process cartridge **102** is attached to the main body of the image forming apparatus **100**, an input/output circuit **201b** of the authentication target chip **103** and the communication circuit **104** of the engine controller **101** are electrically connected to each other. The input/output circuit **201b** relays data that is transmitted and received between the control unit **105** and a microcomputer **202b**. The microcomputer **202b** performs necessary processing in accordance with commands from the control unit **105**. At this time, the microcomputer **202b** stores data necessary for the processing in a volatile memory **203b**. A non-volatile memory **204b** stores a control program **205b** for the authentication target chip **103**, a secret data table **209**, a hash value table **210**, a chip identifier **211**, a combined hash value **213**, a repeat count **214**, and a digital signature **212**.

[0018] The chip identifier **211** is an identifier assigned to each individual authentication target chip **103**, and differs between individual authentication target chips **103**. The secret data table **209** is information (table) indicating a plurality of pieces of secret data for use in authentication. The hash value table **210** is information (table) indicating hash values respectively calculated based on the plurality of pieces of secret data indicated in the secret data table **209**. The method for calculating a hash value in the hash value table **210** based on secret data indicated in the secret data table **209** will be described later. The digital signature **212** is a digital signature (signature information) generated using a signature key based on information indicating the combined hash value **213** and the repeat count **214**. The method of generating the combined hash value **213** will be described later. Note that the signature key used to generate the digital signature **212** corresponds to the signature verification key **215** stored in the authentication chip **106**. In other words, by verifying the digital signature **212** using the signature verification key **215**, it is possible to verify the validity of the combined hash value **213** and the value of the repeat count **214**.

[0019] FIG. 3A is a diagram illustrating the relationship between a plurality of pieces of secret data indicated in the secret data table **209** and a plurality of pieces of hash values indicated in the hash value table **210**. First, M (M is an integer greater than or equal to 2) pieces of secret data sec(1) to sec(M) are generated in a suitable manner, for example, a random manner. The information indicating the M pieces of secret data sec(1) to sec(M) is stored in the non-volatile memory **204b** of the authentication target chip **103** as the secret data table **209**.

[0020] Then, each of the pieces of secret data sec(1) to sec(M) is concatenated with the corresponding chip identifier **211** stored in the non-volatile memory **204b** of the authentication target chip **103** to generate concatenated data. In FIG. 3A, the chip identifier **211** is denoted as “id”. Also, in the following description, “A|B” indicates concatenated data in which data A and data B are concatenated with each other. The concatenated data is then used as an input to the one-way function F to generate a hash value. In FIG. 3A, hsh(m)_1 (m is an integer from 1 to M) indicates a hash value obtained by using concatenated data “id|sec(m)” as an input to the one-way function F. Note that “n” in the notation “hsh(m)_n” indicates the number of times the output of the one-way function F is obtained.

[0021] Then, “hsh(m)_1” is concatenated with the chip identifier **211** to generate concatenated data. In FIG. 3A, “id|hsh(m)_1” indicates concatenated data in which hsh(m)_1 and id are concatenated with each other. Then, the concatenated data is used as an input to the one-way function F to generate a second hash value. In FIG. 3A, hsh(m)_2 indicates a hash value obtained by using “id|hsh(m)_1” as an input to the one-way function F. In this way, by N repeated operations of obtaining the hash value using concatenated data with the chip identifier **211** as an input to the one-way function F, the hash values hsh(1) N to hsh(M) N are obtained based on the respective pieces of secret data sec(1) to sec(M). The information indicating the hash values hsh(1) N to hsh(M) N is stored in the non-volatile memory **204b** of the authentication target chip **103** as the hash value table **210**. Note that in the following description, the M hash values hsh(1) N to hsh(M)_N indicated in the hash value table **210** are simply denoted as hash values hsh(1) to hsh(M).

[0022] Note that the information indicating the repeat count **N** is stored in the non-volatile memory **204b** of the authentication target chip **103** as the repeat count **214** and is used to calculate the digital signature **212**. Note that the one-way function **F** used to obtain the hash value $\text{hsh}(m)$ from the secret data $\text{sec}(m)$ is the same as the one-way function **F** used by the operation unit **207a** of the authentication chip **106**.

[0023] FIG. **3B** is a diagram illustrating the relationship between a plurality of hash values indicated in the hash value table **210** and the combined hash value **213**. As shown in FIG. **3B**, the combined hash value **213** is obtained by using data in which the **M** hash values $\text{hsh}(1)$ to $\text{hsh}(M)$ indicated in the hash value table **210** are concatenated with each other as an input to the one-way function **F**. Note that this one-way function **F** is also the same as that used by the operation unit **207a** of the authentication chip **106**. The combined hash value **213** is stored in the non-volatile memory **204b** of the authentication target chip **103** and is used to calculate the digital signature **212**.

[0024] FIG. **4** is a flowchart of authentication processing executed by the control unit **105**. The control unit **105** starts the authentication processing shown in FIG. **4** in response to the attachment sensor **107** detecting the attachment of the process cartridge **102**. In step **S10**, the control unit **105** reads the chip identifier **211** of the authentication target chip **103**. In step **S11**, the control unit **105** determines whether or not the read chip identifier **211** is stored in the chip identifier table **216**. If the chip identifier **211** is stored in the chip identifier table **216**, the control unit **105** determines that the attached process cartridge **102** has been authenticated. In this case, the control unit **105** determines that the authentication is successful and ends the processing in FIG. **4**.

[0025] On the other hand, if the read chip identifier **211** is not stored in the chip identifier table **216**, the control unit **105** reads, in step **S12**, the digital signature **212**, the combined hash value **213**, and the repeat count **214**, and causes the verifier **206** to verify the digital signature **212**. In step **S13**, the control unit **105** obtains the verification result from the verifier **206**. If the verification result indicates an unsuccessful verification (NG), the control unit **105** determines that the authentication is unsuccessful and ends the processing in FIG. **4**. On the other hand, when the verification result indicating successful verification (OK) is obtained from the verifier **206**, the control unit **105** reads, in step **S14**, the hash values $\text{hsh}(1)$ to $\text{hsh}(M)$ in the hash value table **210**, and causes the operation unit **207a** to calculate the combined hash value. In step **S15**, the control unit **105** causes the comparator **208** to determine whether or not the combined hash value calculated by the operation unit **207a** matches the combined hash value **213** read from the authentication target chip **103**.

[0026] If the control unit **105** receives a comparison result indicating that the two combined hash values do not match from the comparator **208** in **S15**, the control unit **105** determines that the authentication is unsuccessful and ends the processing in FIG. **4**. On the other hand, if the control unit **105** receives a comparison result indicating that the two combined hash values match each other from the comparator **208** in step **S15**, the control unit **105** reads one or more pieces of secret data from the secret data table **209** in **S16**. In the following description, it is assumed that the control unit **105** reads one piece of secret data $\text{sec}(m)$. The control unit **105** causes the operation unit **207a** to calculate the hash value $\text{hsh}(m)$ based on the repeat count **214**, the chip identifier **211**, and the secret data $\text{sec}(m)$.

[0027] The operation unit **207a** obtains the first piece of output data by using input data that is concatenated data of the secret data (**m**) and the chip identifier **211** as an input to the one-way function **F**. Then, the operation unit **207a** obtains the hash value $\text{hsh}(m)$, by performing operations of obtaining the (**k**+1)-th output data using concatenated data of the **k**-th output data and the chip identifier **211** as an input to the one-way function **F**, in a repeated manner from **k**=1 to **N**-1. Note that the value of **N** is indicated by the repeat count **214**.

[0028] In step **S17**, the control unit **105** causes the comparator **208** to determine whether or not the hash value $\text{hsh}(m)$ calculated by the operation unit **207a** matches the hash value $\text{hsh}(m)$ read from the authentication target chip **103**.

[0029] If the control unit **105** receives a comparison result indicating that the two hash values $\text{hsh}(m)$ do not match each other from the comparator **208** in step **S17**, the control unit **105** determines that the authentication is unsuccessful and ends the processing in FIG. **4**. On the other hand, if the control unit **105** receives a comparison result indicating that the two hash values $\text{hsh}(m)$ match each other from the comparator **208** in step **S17**, the control unit **105** determines that the authentication is successful. In this case, the control unit **105** stores the chip identifier **211** read in step **S10** in the chip identifier table **216**.

[0030] Note that in step **S16**, when a plurality of pieces of secret data, for example, K pieces of secret data (K is an integer from **2** to M), are read, the control unit **105** calculates K hash values corresponding to the K pieces of secret data. If all of the calculated K hash values match the corresponding hash values stored in the hash value table **210**, the control unit **105** determines that the authentication is successful. Otherwise, the control unit **105** determines that the authentication is unsuccessful.

[0031] If it is determined that the authentication is unsuccessful, the control unit **105** performs, for example, processing of notifying the user that the process cartridge **102** is not a genuine product.

[0032] By repeating the above-described calculation using the one-way function F multiple times, it is possible to increase the calculation time in the authentication processing without expanding the data length of the secret data $\text{sec}(m)$. This also increases the calculation time required for cryptanalysis attacks such as brute-force attacks, thus improving the resistance to cryptanalysis attacks while reducing an increase in the data length of secret data.

[0033] Note that, in the present embodiment, the concatenated data with the chip identifier **211** is always used as input data to the one-way function F when calculating the hash value from the secret data. However, it is sufficient to use the concatenated data with the chip identifier **211** in at least one of the N calculations using the one-way function F , and it is not necessary to always use the concatenated data with the chip identifier **211** as an input to the one-way function F . For example, in the first calculation using the one-way function F , the concatenated data of the secret data and the chip identifier **211** can be used, and in the second to N -th calculations, the output data of each previous one-way function F can be directly used as an input to the one-way function F . Alternatively, a configuration is also possible in which in the first calculation, the hash value is obtained by using only the secret data as an input to the one-way function F , and in the second to the $(N-1)$ -th calculations, the output data of each previous one-way function F is used as an input, and in the N -th calculation, the concatenated data of the hash value output in the $(N-1)$ -th calculation and the chip identifier **211** is used as an input to the one-way function F . Furthermore, the concatenated data may be used in the odd-numbered operations, and the output data of each previous one-way function F may be used as an input to the one-way function F in the even-numbered operations. Furthermore, the concatenated data can be used in the even-numbered operations, and the output data of each previous one-way function F can be used as an input to the one-way function F in the odd-numbered operations.

[0034] In the present embodiment, the combined hash value **213** and the repeat count **214** serving as plain text, and the digital signature **212** thereof are stored in the non-volatile memory **204b** of the authentication target chip **103**. However, a configuration is also possible in which a digital signature of a message recovery type can be stored as the digital signature **212**. In this case, in step **S12**, the verifier **206** verifies the digital signature **212** and restores the combined hash value **213** and the repeat count **214** from the digital signature **212**.

Second Embodiment

[0035] Next, a second embodiment will be described focusing on the differences from the first embodiment. FIG. **5** is a diagram showing a configuration of an authentication system according to the present embodiment. Compared to the authentication system of the first embodiment shown in FIG. **2**, the second embodiment differs from the first embodiment in that the non-volatile memory **204b** of the authentication target chip **103** stores a readable count **501**. At the time of factory

shipment, the non-volatile memory **204b** of the authentication target chip **103** stores an initial value of the readable count **501**. The initial value is information indicating an integer greater than 0, and defines the upper limit of the number of times secret data can be read by the control unit **105**. [0036] FIG. **6** is a flowchart of processing executed by the microcomputer **202b** of the authentication target chip **103** according to the present embodiment. The processing shown in FIG. **6** is executed in response to the control unit **105** of the authentication chip **106** instructing the microcomputer **202b** to read secret data. In other words, the processing shown in FIG. **6** is executed in step **S16** in FIG. **4** by the authentication target chip **103**.

[0037] Upon receiving the instruction to read the secret data from the control unit **105**, the microcomputer **202b** determines in step **S20** whether or not the value of the readable count **501** is greater than 0. If the value of the readable count **501** is greater than 0, the microcomputer **202b** performs update such that the value of the readable count **501** is decremented by 1 in step **S21**. The microcomputer **202b** then transmits, in step **S22**, the secret data to be read out in accordance with the instruction, to the control unit **105**. On the other hand, if the value of the readable count **501** is not greater than 0 in step **S20**, the microcomputer **202b** notifies, in step **S23**, the control unit **105** that the secret data is not to be transmitted. Upon being notified by the microcomputer **202b** of the fact that the secret data is not to be transmitted, the control unit **105** determines that the authentication is unsuccessful and ends the processing in FIG. **4**.

[0038] Note that in the present embodiment, even if a plurality of pieces of secret data are to be read out in accordance with the instruction from the control unit **105**, the value of the readable count **501** is updated so as to be decremented by 1 in step **S21**. However, a configuration is also possible in which, in step **S21**, the value of the readable count **501** is decremented by the number of pieces of secret data to be read out in accordance with the instruction from the control unit **105**. In this case, the determination in step **S20** is to determine whether or not the value of the readable count **501** is not less than the number of pieces of secret data to be read out in accordance with the instruction from the control unit **105**.

[0039] Also in the present embodiment, if the value of the readable count **501** is not greater than 0, the microcomputer **202b** notifies, in step **S23**, the control unit **105** that the secret data is not to be transmitted. However, a configuration is also possible in which if the value of the readable count **501** is not greater than 0, the microcomputer **202b** does not give any notification to the control unit **105**. In this case, if the microcomputer **202b** does not transmit secret data even after a predetermined period of time has elapsed since the reception of the instruction to read the secret data, the control unit **105** determines that the authentication is unsuccessful.

[0040] According to the present embodiment, as described above, the number of times secret data can be read out is limited. This makes simple data copying difficult and improves security against cryptanalysis attacks.

Other Configurations

[0041] Although the present disclosure has been described with reference to embodiments in which the authentication apparatus is provided in the main body of the image forming apparatus and the authentication target apparatus is provided in the replacement unit of the image forming apparatus, the present disclosure is applicable even to an apparatus different from such an image forming apparatus. In other words, the present disclosure is applicable to any apparatus that uses a replacement unit configured to be detachable from the main body of the apparatus. Note here that the authentication apparatus is attached to the main body of the apparatus, and the authentication target apparatus is attached to the replacement unit of the apparatus.

Other Embodiments

[0042] Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the

above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0043] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0044] This application claims the benefit of Japanese Patent Application No. 2024-031611, filed Mar. 1, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. An authentication apparatus that authenticates an authentication target apparatus, comprising: an acquisition unit configured to acquire, from the authentication target apparatus, information indicating secret data stored in the authentication target apparatus, a first hash value that corresponds to the secret data, and a repeat count N, wherein N is an integer of 2 or more; an operation unit configured to obtain a second hash value from input data that is based on the secret data, by performing N repeated operations using a one-way function; and an authentication unit configured to authenticate the authentication target apparatus by comparing the first hash value acquired by the acquisition unit from the authentication target apparatus with the second hash value obtained by the operation unit.
2. The authentication apparatus according to claim 1, wherein the authentication unit determines that authentication of the authentication target apparatus is unsuccessful if the first hash value does not match the second hash value.
3. The authentication apparatus according to claim 1, wherein the acquisition unit is further configured to acquire, from the authentication target apparatus, signature information for verifying the information indicating the repeat count, and the authentication unit determines that authentication of the authentication target apparatus is unsuccessful if verification using the signature information is unsuccessful.
4. The authentication apparatus according to claim 3, wherein the authentication target apparatus stores a plurality of pieces of secret data and a plurality of first hash values that correspond to the plurality of pieces of secret data, and the signature information is information for verifying the information indicating the repeat count, and information indicating a third hash value that is obtained, using the one-way function, based on a value in which the plurality of first hash values are concatenated with each other.
5. The authentication apparatus according to claim 4, wherein the acquisition unit is further configured to acquire the plurality of first hash values and the third hash value from the authentication target apparatus, the operation unit is configured to obtain a fourth hash value, using the one-way function, based on a value in which the plurality of first hash values are concatenated

- with each other, and the authentication unit determines that authentication of the authentication target apparatus is unsuccessful if the third hash value acquired by the acquisition unit from the authentication target apparatus does not match the fourth hash value obtained by the operation unit.
- 6.** The authentication apparatus according to claim 1, wherein the acquisition unit is further configured to acquire an identifier of the authentication target apparatus from the authentication target apparatus, and the input data is concatenated data of the secret data and data indicating the identifier.
- 7.** The authentication apparatus according to claim 6, wherein the operation unit is configured to obtain the second hash value, by obtaining first output data using the input data as an input to the one-way function, and obtaining $(k+1)$ -th output data using concatenated data of k -th output data and the data indicating the identifier as an input to the one-way function in a repeated manner from $k=1$ to $N-1$.
- 8.** An image forming apparatus that comprises an authentication apparatus, and from which a unit is detachable, the unit including an authentication target apparatus to be authenticated by the authentication apparatus, wherein the authentication apparatus includes: an acquisition unit configured to acquire, from the authentication target apparatus, information indicating secret data stored in the authentication target apparatus, a first hash value that corresponds to the secret data, and a repeat count N , wherein N is an integer of 2 or more; an operation unit configured to obtain a second hash value from input data that is based on the secret data, by performing N repeated operations using a one-way function; and an authentication unit configured to authenticate the authentication target apparatus by comparing the first hash value acquired by the acquisition unit from the authentication target apparatus with the second hash value obtained by the operation unit.
- 9.** An authentication target apparatus that is to be authenticated by an authentication apparatus, comprising: a storage unit configured to store information indicating a plurality of pieces of secret data, first hash values that respectively correspond to the plurality of pieces of secret data, and a repeat count N , wherein N is an integer of 2 or more; and a processing unit configured to perform processing of transmitting the information stored in the storage unit to the authentication apparatus in accordance with an instruction from the authentication apparatus, wherein a first hash value corresponding to secret data of the plurality of pieces of secret data is obtained from input data that is based on the secret data, by N repeated operations using a one-way function.
- 10.** The authentication target apparatus according to claim 9, wherein the storage unit further stores signature information for verifying the information indicating the repeat count.
- 11.** The authentication target apparatus according to claim 10, wherein the signature information is information for verifying the information indicating the repeat count, and information indicating a third hash value that is obtained, using the one-way function, based on a value in which the plurality of first hash values respectively corresponding to the plurality of pieces of secret data are concatenated with each other.
- 12.** The authentication target apparatus according to claim 9, wherein the input data is concatenated data of the secret data and data indicating an identifier of the authentication target apparatus.
- 13.** The authentication target apparatus according to claim 12, wherein the first hash value corresponding to the secret data is obtained, by obtaining first output data using the input data as an input to the one-way function, and obtaining $(k+1)$ -th output data using concatenated data of k -th output data and the data indicating the identifier as an input to the one-way function in a repeated manner from $k=1$ to $N-1$.
- 14.** The authentication target apparatus according to claim 9, wherein, if a number of times of reading the plurality of pieces of secret data by the authentication apparatus reaches an upper limit, the processing unit is further configured not to respond to an instruction to read out secret data from the authentication apparatus or to notify the authentication apparatus that the secret data is not to be transmitted.
- 15.** A replacement unit for an image forming apparatus, comprising an authentication target

apparatus that is to be authenticated by an authentication apparatus of the image forming apparatus, wherein the authentication target apparatus includes: a storage unit configured to store information indicating a plurality of pieces of secret data, first hash values that respectively correspond to the plurality of pieces of secret data, and a repeat count N , wherein N is an integer of 2 or more; and a processing unit configured to perform processing of transmitting the information stored in the storage unit to the authentication apparatus in accordance with an instruction from the authentication apparatus, and the a first hash value corresponding to secret data of the plurality of pieces of secret data is obtained from input data that is based on the secret data, by N repeated operations using a one-way function.
